

Dr. Yiwen Fang

Postdoctoral Researcher, Zhejiang University - University of Illinois Urbana-Champaign Institute, ZJUI

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Research Focus

My research advances the monitoring of surface water resources by combining satellite remote sensing, physics-based modeling, and LLM-powered scientific computing. The foundation of my work is high-resolution mountain snow water estimation through Bayesian data assimilation. I develop sub-kilometer mountain snow water datasets across the western United States, High Mountain Asia, and the Altay mountain based on modeled estimates constrained by Landsat or MODIS retrievals. Building on that foundation, my current work develops AI-enabled scientific computing systems, the Remote Sensing Hub (RSHub) and PhysEarth-Agent, that make physics-based Earth models accessible through natural language while keeping their scientific basis traceable and their results verifiable. My independent research agenda centers on LLM agents for land surface modeling workflow automation, and extends to AI-enabled terrestrial water resources monitoring from emerging satellite observations and to water-related socioeconomic risk assessment.

Education

University of California, Los Angeles (UCLA)	Los Angeles, USA
Ph.D. in Civil Engineering	09/2018 - 03/2023
◦ <i>Dissertation title</i>: Examination of the space-time variability and uncertainty of snow water storage over the Western U.S. and Andes ◦ <i>Committee</i>: William W. Yeh, Dennis P. Lettenmaier, Timu Gallien, Steven A. Margulis (Chair)	
University of California, Los Angeles (UCLA)	Los Angeles, USA
M.S. in Civil Engineering	09/2017 - 07/2018
Hohai University	Nanjing, China
B.S. in Hydrology and Water Resources Engineering	09/2013 - 07/2017

Work Experience

Postdoctoral Researcher (Current Position)	10/2023 - Present, Zhejiang University, China
Software Engineering Intern	04/2023 - 08/2023, Cadence Design Systems, Inc, USA
◦ Developed LLM-based applications for internal documents Q&A and problem detection	
Research Assistant	09/2017 - 04/2023, UCLA, USA

Active Projects

Young Scientists Fund-Type C, National Natural Science Foundation	
青年科学基金项目 (C类)	01/2027 - 12/2029
Data assimilation and uncertainty analysis of snow water equivalent constrained by InSAR phase difference information in alpine mountainous regions, Role: project leader	
National Key Research and Development Program of China (国家重点研发计划)	10/2022 - Present
Research on Full-Spectrum Physical Modeling and Retrieval Algorithms for Snow, PI: Zhenzhan Wang, Role: project contributor	
State Key Laboratory of Remote Sensing Science	10/2025 - Present
Electromagnetic Radiation and Phase Characteristics of Wet Snow under Multi-Frequency Microwave Observations and Sensitivity Analysis of Physical Parameters, PI: Shurun Tan, Role: project contributor	
International Campus, Zhejiang University	09/2025 - Present
Development of an AI-Enabled Virtual Reality Teaching and Experimental Platform for Electromagnetic Fields and Waves, PI: Shurun Tan, Role: proposal contributor	

Past Completed Projects

Chaoyong Program

2024-2025

Snow water equivalent estimation from remote sensing observations, PI: Yiwen Fang

NASA Interdisciplinary Research in Earth Science (IDS)

2020-2023

Snow: Learning how to Scale Success from Field Experiment to the Earth System, PI: Jessica Lundquist, Role: student contributor

National Oceanic and Atmospheric Administration (NOAA) OAR/OWAQ Observations Program

2018-2020

Evaluation and diagnosis of National Water Model Simulations over CONUS using a novel snow reanalysis dataset, PI: Konstantinos Andreadis, Role: student contributor

Amazon Web Service (AWS)

2020

Toward a Global Snow Dataset Over the Remote Sensing Record: A Prototype Case Study over the Western U.S. Using Amazon Web Services, PI: Steve Margulis, Role: proposal and student contributor

Selected Research Projects at ZJUI

PhysEarth-Agent: An LLM Agent for Physics-Based Earth Model Research

- Designed and led development of an open-source LLM agent that plans, configures, executes, and reports physics-based Earth surface modeling studies from a plain-language research goal.
- Released open source and as a hosted browser deployment; selected as a Top-8 finalist in the APGARSS 2026 GeoAI Challenge.

Large Language Model (LLM) Powered Remote Sensing Hub (RSHub)

- Led a team of undergraduate and graduate students to develop an LLM-based cloud computing platform unifying soil, vegetation, and snow microwave scattering models, with direct implications for satellite sensor design and retrieval algorithm development.
- Publicly released at rshub.zju.edu.cn and documented in *IEEE Geoscience and Remote Sensing Magazine* (2025); a follow-on autonomous-agent version (RSHub 2.0) is under review.

Altay Snow Data Assimilation

- Participate in the National Key Research and Development Program on snow data assimilation in Altay, Xinjiang;
- Conducted 2025 snow-season field experiments in the Altay region and generated multi-year 150 m snow water equivalent (SWE) datasets through data assimilation.

AI-Enabled Virtual Reality Teaching Platform for Electromagnetic Fields and Waves Lab

- Collaborating with the lab lecturer to develop the architectural framework for an AI-enabled VR teaching and experimental platform for Electromagnetic Fields and Waves, integrating large language models for intelligent query and interaction.

Grants and Honors

Finalist (Top 8), GeoAI Challenge, APGARSS 2026

10/2026, Qingdao, China

PhysEarth-Agent: an LLM agent for reproducible research with physics-based Earth surface models

Overseas Postdoctoral Talent Introduction Program (博士后海外引才专项)

2024-2026

National Funded Postdoctoral Researcher Program, Category C (国家资助博士后研究人员计划 C档)

2024

University Fellowship, UCLA

2019-2023

U.S./China Clean Energy Research Center for Water-Energy Technologies (CERC-WET) Fellow for Outstanding Contributions to Research

Submitted journal paper under review

- Fang, Y.**, Pan, J., Margulis, S. A., Zhou, J., Sun, H., Lei, Y., Xiong, C., Wang, H., Tan, S. (2026). Assessing the use of satellite products from multispectral unmixing and differential interferometry on Altay snow water equivalent estimation. Submitted to *The Cryosphere*.
- Lin, J., **Fang, Y.**, Chen, K., Zhang, Yao., Huang, S., Liu, H., Zhu, X., Tan, S. (2026). RSHub 2.0: An LLM-based Autonomous Agent for Geophysical Microwave Scattering Modeling. Submitted to *IEEE Geoscience and Remote Sensing Magazine*

Publications

1. **Fang, Y.**, Chen, K., Bai, X., Cao, Y., Du, J., Haider, I., ... & Tan, S. (2025). RSHub: A remote sensing hub for geophysical microwave scattering modeling. *IEEE Geoscience and Remote Sensing Magazine*, 13(4), 406-413.
2. Sun, H., **Fang, Y.**, Margulis, S. A., Mortimer, C., Mudryk, L., & Derksen, C. (2025). Evaluation of the Snow Climate Change Initiative (Snow CCI) snow-covered area product within a mountain snow water equivalent reanalysis. *The Cryosphere*, 19(6), 2017-2036.
3. Du, J., Cao, Y., **Fang, Y.**, Cao, Z. and Tan, S. (2025). Electromagnetic Modeling and Validation of Wet Snow for Passive and Active Microwave Observations: Insights from the NoSREx Dataset and the Altay Field Campaign. In *IGARSS 2025-2025 IEEE International Geoscience and Remote Sensing Symposium*, pp. 920-923, IEEE.
4. **Fang, Y.**, Chen, K., Bai, X., Cao, Y. and Tan, S. (2024). A shared computing platform for remote sensing community: The framework setup and user interface. In *IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium*, pp. 3066-3069, IEEE.
5. Cao, Y., Luo, C., Tan, S., Kang, D. H., **Fang, Y.**, & Pan, J. (2024). A Snow Water Equivalent Retrieval Framework Coupling 1D Hydrology and Passive Microwave Radiative Transfer Models. *Remote Sensing*, 16(10), 1732.
6. **Fang, Y.**, Liu, Y., Li, D., Sun, H., & Margulis, S. A. (2023). Spatiotemporal snow water storage uncertainty in the midlatitude American Cordillera. *The Cryosphere*, 17(12), 5175-5195.
7. Pflug, J. M., **Fang, Y.**, Margulis, S. A., & Livneh, B. (2023). Interactions between thresholds and spatial discretizations of snow: insights from estimates of wolverine denning habitat in the Colorado Rocky Mountains. *Hydrology and Earth System Sciences*, 27(14), 2747-2762.
8. Shulgina, T., Gershunov, A., Hatchett, B.J., Guirguis, K., Subramanian, A.C., Margulis, S.A., **Fang, Y.**, Cayan, D.R., Pierce, D.W., Dettinger, M. and Anderson, M.L. (2023). Observed and projected changes in snow accumulation and snowline in California's snowy mountains. *Climate Dynamics*, 61(9), pp.4809-4824.
9. Ban, Z., Xin, C., **Fang, Y.**, Ma, X., Li, D., & Lettenmaier, D. P. (2023). Snowmelt-radiation feedback impact on western US Streamflow. *Geophysical Research Letters*, 50(23), e2023GL105118.
10. Ma, X., Li, D., **Fang, Y.**, Margulis, S. A., & Lettenmaier, D. P. (2023). Estimating spatiotemporally continuous snow water equivalent from intermittent satellite observations: An evaluation using synthetic data. *Hydrology and Earth System Sciences*, 27(1), 21-38.
11. **Fang, Y.**, Liu, Y., & Margulis, S. A. (2022). A western United States snow reanalysis dataset over the Landsat era from water years 1985 to 2021. *Scientific Data*, 9(1), 677.
12. Liu, Y., **Fang, Y.**, & Margulis, S. A. (2021). Spatiotemporal distribution of seasonal snow water equivalent in High-Mountain Asia from an 18-year Landsat-MODIS era snow reanalysis dataset. *The Cryosphere*, 15(11), 5261–5280.
13. Liu, Y., **Fang, Y.**, Li, D. and Margulis, S.A., (2022). How well do global snow products characterize snow storage in High Mountain Asia?. *Geophysical Research Letters*, 49(16), p.e2022GL100082.
14. Margulis, S. A., **Fang, Y.**, Li, D., Lettenmaier, D. P., & Andreadis, K. (2019). The utility of infrequent snow depth images for deriving continuous space-time estimates of seasonal snow water equivalent. *Geophysical Research Letters*, 46(10), 5331-5340.

Conference Talks (first author only)

1. **Fang, Y.**, Tan, S. (2026, July). Daily Subkilometer Altay Snow Water Equivalent Estimation via Assimilation of Harmonized Landsat Sentinel Fractional Snow Covered Area. In *PIERS 2026*. Suzhou.
2. **Fang, Y.** (2025, November). Multi-source remote sensing snow data assimilation in the Altay mountains. In *2nd Zhejiang Frontier Interdisciplinary Forum on Remote Sensing, 2025*. Hangzhou Normal University.
3. **Fang, Y.** (2025, May). Remote sensing computing platform based on microwave scattering models. In *5th National Radar Earth Observation Conference, 2025*. Nanjing University, Suzhou Campus.
4. **Fang, Y.**, Cao, Y., Tan, S., Kang, D. H. (2024, April). Snow Water Equivalent Retrieval from Coupled Radiative Transfer and Hydrology Models by Assimilating Active and Passive Microwave Observations. In *PIERS 2024*. Chengdu.
5. **Fang, Y.**, Chen, K., Bai, X., Cao, Y., Tan, S. (2024, July). A Shared Computing Platform for Remote Sensing Community: The Framework Setup and User Interface. In *IGARSS 2024*. Athens, Greece.
6. **Fang, Y.**, Liu, Y., & Margulis, S. A. (2020, December). A New Landsat-era Snow Reanalysis Dataset over the Western United States. In *AGU Fall Meeting 2020*. AGU.

7. **Fang, Y.** (2020, October) Characterizing snowpack and snowmelt for water and hydropower management in California. *In Water-Energy Technologies U.S.-China Clean Energy Research Center (CERC-WET)*. Berkeley, California.
8. **Fang, Y.**, Margulis, S. A., & Liu, Y. (2019, December). Exploring the impact of explicit assimilation of dust-on snow albedo information on snow estimates from a snow reanalysis framework. *In AGU Fall Meeting 2019*. AGU.

Datasets

1. **Fang, Y.**, Liu, Y., & Margulis, S. A. 2022. Western United States UCLA Daily Snow Reanalysis, Version 1. Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. <https://doi.org/10.5067/PP7T2GBI52I2>
2. Liu, Y., **Fang, Y.**, Margulis, S. A. 2021. High Mountain Asia UCLA Daily Snow Reanalysis, Version 1. Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. <https://doi.org/10.5067/HNAUGJQXSCVU>

Teaching Experience

Invited Lecture	2024, Zhejiang University
◦ Frontier Engineering Technology Lecture Series: Basics of Snow Remote Sensing and Data Assimilation	
Teaching Assistant	2018 - 2022, UCLA
◦ Applied Numerical Analysis & Modeling Introduction to Hydrology	
Student Advising	
Summer Student Program	Summer 2025, Zhejiang University
◦ Advised undergraduate students, Jiaming Lin, Shenghong Huang, Yao Zhang, Hao Liu, Xiyuan Zhu on LLM agents for geophysical modeling; the work produced the RSHub 2.0 manuscript (under review) and PhysEarth-Agent, an APGARSS 2026 GeoAI Challenge Top-8 finalist	
Undergraduate Thesis	Spring 2025, Zhejiang University
◦ Advised undergraduate student, Qiushi Liu for his undergraduate thesis	
RSHub Student Project	2024-2025, Zhejiang University
◦ Advised undergraduate students, Jiayang Chen, Richie Tjuahdi, Davidson Leonard for student project	
◦ Advised graduate students, Yuanhao Cao, Xuyang Bai, Kaiqi Chen, Jiayi Du, for research project	
C&EE 199: Directed Research in Civil Engineering	Spring 2019, UCLA
◦ Advised senior undergraduate student, Lucia Pan for her research project	

Patents and Software Copyrights (Under Review)

1. Large Language Model based parameter tuning approach and system. Campus # ZL-2026-1-00776
2. Remote Sensing Hub for Microwave Scattering Models. Campus # ZZQ-2026-00229

Academic Services

Manuscript review
Reviewed manuscripts for journals including:
◦ Scientific Reports; Theoretical and Applied Climatology; Hydrology and Earth System Sciences; Water Resources Research; Journal of Hydrology; Journal of Hydrology: Regional Studies; Journal of Hydrometeorology